



Human Resource Information System (HRIS) and Timekeeping System

Prepared for: Prospective Third-Party Software / Web Development Contractors (Bidders)

Prepared by: HR / IT Team

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PART A. PROJECT OVERVIEW

A.1 Objective

Design, develop, customize, implement, and support an integrated Human Resource Information System (HRIS) and Timekeeping System for Philippine Coastal Storage & Pipeline Corporation (PCSPC), covering employee records, leave management, overtime, timekeeping, medical records, training, performance management, compensation and benefits, workflow approvals, reporting, and security administration.

A.2 Scope of Work

The Contractor shall conduct business process review, system design, database architecture, implementation, testing, deployment, training, warranty support, and post-implementation assistance. In addition to the functional scope described in this Part A, the Contractor's engagement is governed by the technical delivery standards set out in Part B of this TOR, including the required DevOps/DevSecOps lifecycle, CI/CD pipeline, source code repository policy, infrastructure environments, database platform and optimization requirements, and documentation and ownership terms. Refer to Annex A HRIS working document.

A.3 Core Functional Modules

- Employee 201 Records: Personal, Family, Education, Employment, Training, and Medical Records
- Leave Management
- Overtime and OT Allowance
- Timekeeping Management and Computation
- Biometric Machine Integration to Timekeeping Management System.
- Performance Management
- Training and Development
- Compensation and Benefits
- Travel History
- Loans Management
- Deduction Management
- Earnings Management
- Medical Management
- Ot Report generation Summary
- Reports and Analytics



A.4 Enhanced Requirements from HRIS Working Document

- Web Access
- Mobile accessibility
- Multi-Factor Authentication (MFA)
- Role-based security
- Audit logs
- Configurable approval workflows
- Document repository
- Employee self-service
- Holiday calendar management
- Training confirmation
- Medical reimbursement management
- Historical employee movement tracking
- Customizable reports
- Automated notifications
- Backup and recovery
- API integration support

A.5 Employee Information Requirements

The system shall maintain complete employee master data, including employee number, personal information, dependents, emergency contacts, statutory numbers, compensation history, personnel movements, education, work experience, training history, and performance records.

A.6 Approval Workflow Requirements

The system shall support configurable multi-level approval workflows for Leave, Overtime, OT Meal Allowance, Travel Requests, and other HR transactions, following PCSPC's departmental approval matrices.

A.7 Security Requirements

The system shall implement role-based access control, MFA, password policy management, secure database configuration, encryption of sensitive information, audit trails, activity logs, and account lockout protection, within a cybersecurity-compliant architecture. Detailed, implementation-level DevSecOps controls that operationalize this requirement throughout the development lifecycle are set out in Section B.9.



A.8 Reporting Requirements

The system shall provide employee reports, leave reports, overtime reports, training reports, medical reports, deductions and earnings reports, attendance summaries, manhour reports, and customizable exports to Excel.

A.9 Contractor Responsibilities

- Provide a project plan and conduct requirements workshops.
- Recommend appropriate hardware and software.
- Perform testing and ensure the system is free of bugs and vulnerabilities to avoid hacking incidents.
- Fix and debug defects in the event of system errors or glitches.
- Provide user manuals and conduct training.
- Maintain a readable source code repository and turn it over to PCSPC.
- Provide warranty and support.
- Comply with the technical delivery standards defined in Part B, including the DevOps/DevSecOps lifecycle, CI/CD pipeline, source code and versioning policy, and documentation requirements.
- Prepare and submit weekly progress reports.
- Assign a dedicated Project Manager.
- Conduct knowledge transfer to PCSPC IT personnel.
- Correct all defects identified during warranty.

A.10 PCSPC Responsibilities

- Assign a Project Focal Person.
- Provide process information, business rules, and approval workflows.
- Provide attendance computation requirements.
- Conduct timely review of deliverables.
- Provide and maintain the infrastructure environments, official source code repository, and cloud resources described in Part B.

A.11 Deliverables

- System Design Document
- Database Design
- Configured HRIS Application
- User Manual
- Training Materials



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- Test Results
- Warranty Support
- Complete, Readable Source Code, Database Scripts, Configuration Files, and Repository Access
- Full documentation set described in Section B.7 (architecture, logic flow, setup guide, dependencies, deployment runbook, and administrator manual)

A.12 Contract Duration

Project duration shall be proposed by the bidder and approved by PCSPC.

PART B. TECHNICAL DELIVERY STANDARDS: DEVOPS, DEVSECOPS, CI/CD, INFRASTRUCTURE, AND DATA

B.1 Software Development Life Cycle: DevOps and DevSecOps Requirements

The Contractor must follow a documented DevOps and DevSecOps life cycle for all phases of development. This means security, quality checks, and operational readiness must be built into every stage of development, not treated as a final step before launch.

Stage	Requirement
Plan	Requirements, user stories, and technical designs are documented and version-controlled before development begins.
Code	Development follows agreed coding standards, with peer/code review prior to merge into shared branches.
Build	Automated build processes compile/package the application consistently across environments.
Test	Automated and manual testing (unit, integration, security, and user acceptance testing) is performed before promotion to the next environment.



Release	Releases are versioned, approved, and traceable to a specific, tagged source code commit.
Deploy	Deployment to Dev, Staging (UAT), and Live follows a controlled, repeatable, and documented process.
Operate	The application is monitored in Live for availability, errors, and performance.
Monitor / Secure	Continuous security monitoring, vulnerability scanning, and access reviews are performed (DevSecOps).

B.2 CI/CD Pipeline Requirements

The Contractor must implement a proper Continuous Integration (CI) and Continuous Delivery (CD) pipeline for the HRIS and Timekeeping application. At minimum, the CI/CD pipeline must include:

- **Continuous Integration:** Every code change pushed to the repository is automatically built and tested, with results visible to both the Contractor's team and PCSPC.
- **Automated Testing Gates:** Code must pass automated tests (unit tests, and where applicable, security and code-quality scans) before it can be merged or promoted.
- **Continuous Delivery:** Successfully tested builds are automatically prepared and made ready for deployment to Dev and Staging (UAT) environments, with a controlled approval step before deployment to Live.
- **Environment Promotion:** Code must be promoted through Dev → Staging (UAT) → Live in that order. No change may be deployed directly to Live without first passing through Dev and Staging (UAT).
- **Pipeline Visibility:** PCSPC must be given access (read-only, at minimum) to the CI/CD pipeline dashboard/logs to monitor build and deployment status.
- **Rollback Capability:** The pipeline must support rolling back to a previous stable version in case a deployment introduces a defect.

B.3 Source Code Repository, Branching, and Versioning Policy

B.3.1 Official Repository

PCSPC will provide and own the official source code repository for the project (e.g., on GitHub, GitLab, Bitbucket, or an equivalent platform, or self-hosted, as decided by PCSPC). All development work must be committed to this official repository. The Contractor must not maintain the authoritative copy of the source code on its own private infrastructure.



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B.3.2 Push and Merge Requirements

- The Contractor must follow an agreed branching strategy (e.g., a main/production branch, a staging branch, and feature/development branches) to support safe, traceable deployment.
- All code must be pushed to the official repository regularly (at minimum, at the end of each working day or development milestone, whichever is more frequent).
- Merges into the staging and production branches must go through a pull/merge request with at least one review and passing CI checks before approval.
- Direct, unreviewed pushes to the main/production branch are not permitted.
- Every merge to staging or production must correspond to a specific, documented deployment to that environment.

B.3.3 Version Retention

PCSPC must have the complete source code for every version/release pushed to the repository, including historical versions, to support documentation, audit, and source code tracing. Tags or releases must be created in the repository for every version deployed to Staging (UAT) or Live, clearly labeled with a version number and date.

B.4 Infrastructure and Environments (Dev, Staging/UAT, Live)

Environment	Purpose	Access
Development (Dev)	Active development and initial testing by the Contractor.	Contractor (primary), PCSPC (read/monitoring access)
Staging (User Acceptance / UAT)	Pre-production testing, user acceptance testing by PCSPC, and validation before go-live.	Contractor and PCSPC (shared)
Live (Production)	The system used by actual PCSPC end users.	Restricted; deployment only through the approved CI/CD process

B.5 Database Requirements (PostgreSQL) and Performance Optimization

B.5.1 Database Platform



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PostgreSQL shall be the primary database management system for the HRIS and Timekeeping application. The database shall be deployed on a separate, dedicated server from the application server(s), consistent with standard three-tier architecture practices.

B.5.2 Performance and Optimization Requirements

To minimize query bottlenecks once the application is live, the Contractor must, at minimum:

- Design a normalized (or appropriately denormalized for performance, with justification) database schema based on the approved data model.
- Apply proper indexing strategies for frequently queried fields and foreign keys.
- Optimize queries, including avoidance of N+1 query patterns and use of query plans (EXPLAIN/ANALYZE) to validate performance for high-traffic operations such as timekeeping and biometric data processing.
- Implement connection pooling appropriate to expected concurrent load.
- Implement caching strategies where appropriate for frequently accessed, rarely changing data.
- Provide a database backup and recovery plan, including backup frequency and restore procedures.
- Conduct load/performance testing prior to Live deployment and provide PCSPC with the results.

B.6 Cloud Infrastructure Provisioning (IaaS)

PCSPC will provide the Infrastructure as a Cloud (IaaS) environment for the project, using Google Cloud Platform (GCP) or another leading cloud provider as determined by PCSPC. The Contractor is responsible for specifying the infrastructure requirements needed (e.g., compute, storage, networking, and database service tiers) to support the application across Dev, Staging (UAT), and Live, and for working with PCSPC's designated cloud administrator(s) to provision and configure these resources correctly.

B.7 Documentation Requirements

The Contractor must provide complete, professional documentation as part of project deliverables, including at minimum:

- **System Architecture Documentation:** overall structure of the application, technology stack, and how components interact.
- **Logic Flow Documentation:** how key business processes and workflows (e.g., leave approval, overtime computation, timekeeping/biometric processing) operate within the system.
- **Setup and Installation Guide:** step-by-step instructions to install, configure, and run the application in each environment.
- **Dependencies and Add-ons List:** all libraries, frameworks, third-party services, and packages used, including versions and licensing.



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- **Database Schema Documentation:** entity-relationship diagrams and table/field descriptions.
- **API Documentation:** endpoints, request/response formats, and authentication methods, including any biometric device or third-party integration APIs.
- **Deployment Runbook:** the exact steps to deploy a new release to each environment.
- **User Manual:** for end users of the HRIS and Timekeeping system.
- **Administrator Manual:** for PCSPC's IT staff to maintain and support the system after handover.

All documentation must be delivered in editable, non-proprietary formats (e.g., Markdown, Word, or equivalent) and kept up to date with each major release. This requirement operationalizes the User Manual and documentation deliverables already listed in Section A.11.

B.8 Source Code Ownership and Intellectual Property

- All source code developed under this engagement is the sole property of PCSPC.
- The source code delivered and stored in the official repository must be human-readable and properly commented. Obfuscated, minified-only, or otherwise deliberately obscured source code is not acceptable as a final deliverable, consistent with the "readable source code repository" hand-over already required under Section A.9.
- Any third-party libraries or components used must be clearly identified, with license terms compatible with PCSPC's ownership and use of the system.
- Upon contract completion or termination, the Contractor must transfer all credentials, access, and remaining project artifacts to PCSPC.

B.9 DevSecOps Security Controls (Supplements A.7)

In addition to the security requirements already stated in Section A.7 (role-based access control, MFA, password policy, encryption, audit trails, and account lockout protection), the Contractor must implement the following controls throughout the development lifecycle:

- Secrets (passwords, API keys, credentials) must never be committed to the source code repository; a secrets management approach must be used.
- Security scanning (static application security testing (SAST), and where applicable, dynamic testing (DAST)) must be part of the CI/CD pipeline.
- All data in transit must be encrypted (HTTPS/TLS); sensitive data at rest (e.g., medical records, statutory numbers, compensation data) must be encrypted or otherwise protected in accordance with applicable data privacy requirements.
- Access to Staging (UAT) and Live environments and to the source code repository must be limited to authorized personnel, with access logs available for review.
- Regular vulnerability assessment must be performed to ensure the system remains free of exploitable weaknesses, consistent with the Contractor's obligation under Section A.9 to prevent hacking incidents.



B.10 Testing and Quality Assurance

- The Contractor must provide a test plan covering functional, integration, security, performance/load, and user acceptance testing.
- Automated unit and integration tests must be included in the CI pipeline, with a reasonable minimum coverage target agreed upon with PCSPC.
- PCSPC will conduct formal User Acceptance Testing (UAT) in the Staging environment prior to any Live deployment, and the Contractor must support this process by resolving identified defects within an agreed timeframe.

B.11 Deployment and Release Management

- Deployments to Live must occur only after written/formal approval from PCSPC, following successful UAT sign-off.
- Each release must be documented with a release note describing what changed, the associated repository tag/version, and any migration or rollback steps.
- A rollback plan must be defined and tested for each Live deployment.

B.12 Roles and Responsibilities Matrix (Client vs. Contractor)

This matrix operationalizes the responsibilities already identified in Sections A.9 and A.10 for the technical delivery standards defined in this Part B.

Area	PCSPC (Client) Responsibility	Contractor Responsibility
Requirements	Provide business requirements, process information, and approval matrices; validate designs and approve scope changes	Conduct business process review; translate requirements into technical design and confirm feasibility
Source Code Repository	Provide and own the official repository and access management	Push all code to the official repository following the agreed branching policy
Infrastructure	Provide and fund Dev, Staging (UAT), and Live environments and cloud infrastructure (IaaS)	Specify infrastructure requirements and configure the application to run properly on provided infrastructure



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Database	Provide the dedicated database server	Design, build, optimize, and document the PostgreSQL database
CI/CD	Grant necessary access/approvals for deployment gates	Build and maintain the CI/CD pipeline and ensure it is followed for every release
Testing	Perform UAT and provide timely feedback	Perform unit, integration, and pre-UAT testing; fix defects found
Documentation	Review and accept documentation deliverables	Produce and maintain complete, accurate, non-obfuscated documentation
Security	Define data privacy/compliance requirements	Implement DevSecOps controls throughout development and deployment
Clarifications	Respond to Contractor questions in a timely manner	Provide clarification to PCSPC on any technical matter at no additional cost

B.13 Communication, Clarification, and Change Requests

PCSPC reserves the right to request clarification from the Contractor on any aspect of the system, including logic flow, setup, dependencies, architecture, or documentation, at any point during and after development, without any additional charge. The Contractor must respond to reasonable clarification requests within an agreed turnaround time (e.g., one to three business days, to be confirmed with the Contractor).

Any change to the agreed scope of work must be documented as a formal change request, with any impact on cost or timeline agreed upon in writing before work proceeds.

B.14 Support, Warranty, and Maintenance

- The Contractor shall provide the warranty and support referenced in Section A.9, including a warranty period of six (6) months from the date of Live Go-Live or Final Acceptance by PCSPC, whichever is later. During the warranty period, the Contractor shall, at no additional



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cost to PCSPC, correct all defects, bugs, errors, and system malfunctions attributable to the original scope of work, including the provision of necessary patches, fixes, and technical support to ensure the system operates in accordance with the approved specifications.

- The Contractor shall provide a separate quotation for maintenance and technical support services beyond the six (6)-month warranty period. Engagement of such services shall be at the sole discretion of PCSPC and shall be covered by a separate agreement, if applicable.

B.15 Acceptance Criteria and Project Sign-off

- Each phase or milestone deliverable identified in Section A.11 shall be submitted to PCSPC for review and acceptance. PCSPC shall formally accept the deliverable or provide documented comments requiring correction before the Contractor proceeds to the next phase or becomes eligible for the corresponding progress payment.
- Final project acceptance shall be subject to the successful completion of User Acceptance Testing (UAT), submission and approval of all required documentation, turnover of the complete and readable source code together with repository access, and confirmation that the Live Production Environment is fully operational and stable.
- Upon the Contractor's written request for Final Acceptance and verification that all contractual obligations have been satisfactorily fulfilled, PCSPC shall issue a Certificate of Final Acceptance, which shall serve as the basis for final payment, subject to the terms and conditions of the Contract.

B.16 Confidentiality and Data Privacy

The Contractor must treat all PCSPC data, including employee personal, medical, and compensation data processed by the HRIS and Timekeeping system, as strictly confidential, and must comply with the requirements of the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) and its implementing rules, or equivalent applicable data protection regulations. A non-disclosure agreement (NDA), if not already executed, should be signed prior to project commencement.

B.17 Risk Management

The Contractor must proactively identify and communicate project risks (technical, schedule, or resource-related) to PCSPC as soon as they are identified, along with proposed mitigation options.

PART C. GENERAL PROVISIONS

C.1 Interpretation

In the event of any inconsistency between the Contractor's proposal and this Terms of Reference, the provisions of this Terms of Reference shall prevail unless otherwise agreed in writing by PCSPC.



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C.2 Reservation of Rights

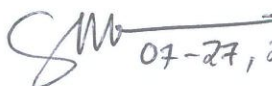
PCSPC reserves the right to reject any or all proposals, waive minor informalities, request clarifications, or cancel the procurement process when deemed to be in its best interest.

End of Terms of Reference

Submission of a proposal shall constitute the Contractor's acknowledgment that it has examined and understood all requirements of this Terms of Reference and has based its proposal accordingly.

Prepared by:

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 07-27, 2024

Position: IT Administrator

Reviewed by:

Name: Ernest Deniega



Position: Senior IT Administrator

Approved by:

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Position: HR/ IT Head